

K1450 — the mistake yielded more strength (Casey’s frame, vindicated this round). (1) ★ DE gate CERTIFIED (Cal §460) and STRENGTHENED (Grace’s no-offset control): BST’s no-crossing is essentially tied with the phantom even with ZERO offset (1630.3 vs 1629.7) — the crossing is barely preferred by the data at all, and the documented low-z offset only tips a near-tie. pred_{125} teeth at definitive scope, honest mild-preference framing ($\sim 1.8\sigma$; “phantom not required, degenerate with a physical systematic”). (2) ★★ THE SIGN QUESTION SETTLED — target-innocently, and by a CLEANER route than Lichnerowicz: the SEA forces D to be Hermitian (a negative-energy spectral projector needs a real spectrum with a “below zero” to fill; anti-Hermitian D has purely imaginary eigenvalues $\rightarrow$ no sea). So $c \geq 0$ and the prediction is +8.75 (Elie 5224). My two-way hermiticity fork was ONE fork in two vocabularies (grading-odd D: Hermitian $\square$ Krein-anti-self-adjoint). (3) The operator still needs the real causal build — the assembled one has no self-adjointness structure at all (fails all four conditions); Lyra removed the stray-i fake and rebuilt the positive operator symmetric-by-construction, blind; the sign gate stays SHUT until the causal operator is built. (4) ★ Elie’s silent-trap catch (bank it): numpy’s **eigvalsh/eigh** *assume* Hermiticity and return plausible-but-wrong real eigenvalues for a non-Hermitian operator with no error — assert Hermiticity before the sea call. Cal cold-reads.

Keeper (2026-08-13, $\sim 12:45$. Casey: “we’ll find mistakes, and even those will show more strength.” This round is the proof — Lyra’s stray-i bug got caught, and fixing it surfaced a deeper truth: the sign isn’t a curvature convention, it’s forced by the sea’s very existence. And Grace’s certification work surfaced a control that strengthened the DE result. Mistakes $\rightarrow$ cleaner truths. Cal cold-reads. Nothing pushed.)

On Casey’s frame — vindicated in one round

“We will find some mistakes, and even those will yield and likely show even more strength for BST.” This round is the clean example, twice over: - The stray-i bug $\rightarrow$ a deeper sign derivation. The sign was about to come from a bookkeeping artifact. Removing it forced the real question, and the answer is *stronger* than the Lichnerowicz argument we had: +8.75 is forced because the Dirac sea cannot exist unless D is Hermitian — a physical necessity, not a curvature convention. A better foundation than the one the bug was hiding. - The certification work $\rightarrow$ a stronger DE result. Grace’s no-offset control (below) shows BST’s no-crossing is *already competitive without any calibration offset* — the win is more robust than the offset-fit alone suggested. The tiers stay honest (the operator isn’t built, the sign is derived-in-principle not yet certified) — but the direction is exactly what you said: the errors are load-bearing, and they’ve been strengthening the thing.

1. ★ DE gate CERTIFIED (Cal §460) + STRENGTHENED (Grace’s control)

- Cal certified the low-z offset was free, not tuned — four checks: fairly counted (+2 params = CPL’s +2, the decisive one); physical magnitude (0.05 mag, within the 0.02–0.05 documented budget, upper edge noted honestly); physically-motivated low-z shape (the phantom residual is low-z-dominated, -13.8 mmag at $z=0.05$ = an anchor-offset signature); BST chose no-crossing freely (tilt A $\rightarrow$ 0 on its own). DE gate cleared:

the crossing is calibration, not cosmology.

- ★ Grace’s no-offset control makes it cleaner still: with the offset fixed to exactly zero, BST’s monotone no-crossing = 1630.3 vs the phantom 1629.7 — a tie ($\Delta\chi^2=0.06$). So BST’s no-crossing *already competes with the phantom without any offset*; the phantom’s preference over no-crossing is a weak $\sim 2\sigma$ pull, and the offset only tips a near-tie. The offset isn’t doing the heavy lifting — the data barely prefer the crossing in the first place. BST wins over a *range* (offset 0–0.06, threshold 0.05–0.15), not a fine-tuned point.
- ★ Honest scope (Cal #27, held on the favorable result): $\Delta\chi^2=3.2$ is a mild $\sim 1.8\sigma$ preference, so the claim is “the phantom crossing is not required — it is degenerate with a physical, documented low- z systematic,” NOT “BST decisively beats the phantom.” At-least-as-well, phantom-not-required, on both direct probes (radial-BAO + SNe). This is the right tier, and it’s a genuine win at it.

2. ★★ THE SIGN — SETTLED by the sea (Elie 5224), target-innocent

- The sea forces Hermiticity. The negative-energy filling is $\chi_{-}\{(-\infty,0)\}(D)$, a spectral projector onto states *below zero* — which requires a real spectrum. A Hermitian D has real eigenvalues; an anti-Hermitian D has purely imaginary ones (verified: $\pm 7.399i$, $|\text{Re}| = 2.7 \times 10^{-15}$). So D must be Hermitian, or the sea does not exist as an object. ¶ branch (a): $c \geq 0$, prediction written +8.75. Target-innocent — it comes from what we’re building (a sea needs a real spectrum), not from anyone preferring a sign. Settles the SIGN only; the magnitude branches (8.50 / 8.75 / 0 / neither) don’t move — no steering of the number Elie’s about to measure.
- My two-way fork was one fork in two vocabularies (Elie): both candidate operators are grading-odd ($\Gamma_5 D \Gamma_5 = -D$ exactly), and for grading-odd D , Hermitian ¶ Krein-anti-self-adjoint. So “drop the i (Hermitian)” and “the Krein-self-adjoint construction” are the same condition — my §1449 “Hermitian or Krein-Hermitian?” was one question. Clarified.

3. The operator still needs the real causal build (sign gate stays SHUT)

- The assembled operator has NO self-adjointness structure — it fails all four conditions (0.56–1.87 on each); only the grading survives. So it’s *stronger* than “mixed hermiticity”: there’s no self-adjointness at all yet, and the fix is real work, not a relabel.
- Lyra removed the fake and rebuilt the positive operator honestly — as an object plus its own mirror image (symmetric by construction, no free symbol to fudge), verified to machine precision, blind (not steered to -8.75). She named the evaluation point (the answer is a *global* property of the whole space, not read off at the flat center where $A=0$).
- ★ Lyra’s own honest call, which I ratify: keep the sign gate SHUT until the causal operator is built. The sign is now derived-in-principle (the sea forces Hermitian $\rightarrow +8.75$) but not yet nailed to the specific causal operator. Next build: the causal operator, blind. *Better a slower true answer than a fast fake one.*

4. ★ BANK — Elie’s silent-trap catch (a standing numerical-safety check)

`numpy.linalg.eigvalsh/eigh` assume Hermiticity and read only one triangle. Fed the current (non-Hermitian) operator, `eigvalsh` returns ± 7.399 real, confidently, with no error — when the truth is ± 7.399 imaginary. And `dolbeault_sea` calls `eigh`. So running the sea on the current operator would silently produce a plausible-looking WRONG sea and raise nothing. ¶ Standing check: assert the structural precondition ($\|D - D^\dagger\| \approx 0$) BEFORE any Hermiticity-assuming numerical routine. This is the numerical sibling of the content-audit gate — a routine that silently returns a plausible-wrong answer is exactly how a result gets faked. Elie also published the measurement instrument in advance (`measure_c`: operator + Lyra’s named point + locked tolerance as declared inputs $\rightarrow$ verdict against the four branches, no discretion) — the checker committed blind. Both banked.

Route

1. ★ Lyra — build the causal operator, blind (the sea version, Hermitian by construction per Elie’s forcing); then the sign is certified, not assumed. Keep the gate shut until it’s done.

2. ★ Elie — the instrument is published + blind; assert Hermiticity before the sea call; measure the instant the causal operator lands + Cal certifies. Still hasn't read c.
3. ★ Cal — sign-cert held (correctly); certify when the causal operator lands (independence + geometry-forced-Hermitian-sign + genuine-Weyl-norm). DE gate CERTIFIED — the crossing is calibration; hold the mild- $\sim 1.8\sigma$ scope.
4. Grace — the strengthened DE result (no-offset tie) into the Falsifiable paper; the definitive T2560 stands, certified.
5. Keeper — ratified the DE win + the sign settlement + Lyra's honest rebuild; banked the eigvalsh-Hermiticity trap + the blind instrument; narrative Axioms pilot drafted (Cal owes the tier-inflation re-read); RETIRE sweep + Guide accuracy-sync continuing; numpy fixed.

— Keeper, K1450, 2026-08-13. Casey frame (“mistakes yield more strength”) VINDICATED: stray-i bug → deeper sign derivation (sea forces Hermitian, better than Lichnerowicz); certification work → stronger DE (no-offset tie). (1) DE GATE CERTIFIED (Cal §460, 4 checks: +2p=CPL fair, 0.05mag in 0.02-0.05 budget, low-z anchor shape, tilt $A \rightarrow 0$ free) + STRENGTHENED (Grace no-offset control: offset=0 → BST 1630.3 vs phantom 1629.7 = TIE $\Delta\chi^2=0.06$; crossing barely preferred, offset only tips near-tie; BST wins over range not point); honest scope $\Delta\chi^2=3.2 \sim 1.8\sigma$ MILD = “phantom not required, degenerate w/ documented systematic” NOT “decisively beats”; crossing = calibration not cosmology, both direct probes. (2) SIGN SETTLED (Elie 5224) target-innocent by CLEANER route than Lichnerowicz: sea = $\chi_{-}\{(-\infty, 0)\}(D)$ needs real spectrum → Hermitian D (real eigenvalues) vs anti-Herm ($\pm 7.399i$ verified) → D MUST be Hermitian or sea doesn't exist → $c \geq 0 \rightarrow +8.75$; settles SIGN only, magnitude branches unmoved; Keeper's two-way fork = ONE fork (grading-odd $\Gamma_5 D \Gamma_5 = -D$: Hermitian $\boxtimes$ Krein-anti-self-adjoint). (3) Operator needs real causal build: assembled fails ALL 4 self-adjointness (0.56-1.87), only grading survives = no self-adjointness at all, fix is real work; Lyra removed fake, rebuilt positive operator symmetric-by-construction (object+mirror) blind machine-precision, named eval point (global not flat-center); SIGN GATE STAYS SHUT till causal operator (sign derived-in-principle not nailed). (4) BANK Elie silent-trap: eigvalsh/eigh ASSUME Hermiticity, return plausible-wrong REAL eigenvalues for non-Herm operator NO error; dolbeault_sea calls eigh → running sea now = silent plausible-wrong sea → ASSERT $\|D - D^\dagger\| \approx 0$ before Herm-assuming routine (numerical sibling of content-audit gate); Elie published measure_c instrument in advance (no discretion, checker blind). Route: Lyra build causal operator blind (Herm by construction); Elie assert-Herm + measure when causal lands + certified; Cal sign-cert held+DE-certified hold- 1.8σ -scope; Grace strengthened-DE into paper; Keeper ratified + banked trap+instrument + narrative pilot drafted. Cal cold-reads. Nothing pushed.